Bank of Leeds Public Policy Proposal

Sheffield Victoria–Manchester 125

A New High-Speed Trans-Pennine Railway via the Woodhead Corridor

Prepared for: HM Treasury • Department for Transport • South Yorkshire Mayoral Combined Authority • Greater Manchester Combined Authority

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Executive Summary

This proposal examines a **new-build high-speed railway** connecting a **reopened Sheffield Victoria Station** with Manchester using the historic Woodhead corridor as the basis for a modern 125 mph (200 km/h) railway.

Unlike upgrading the existing Hope Valley Line—which is constrained by the geography of the Peak District—this proposal argues that a largely new alignment would provide:

- significantly faster journey times,
- dedicated passenger capacity,
- greater freight resilience on existing routes,
- improved regional economic integration.

It is important to note that **this is a speculative infrastructure proposal**. Current UK Government policy is focused on upgrading the Hope Valley corridor and Northern Powerhouse Rail using existing routes. The Government's Integrated Rail Plan explicitly considered, but rejected, a new Sheffield–Manchester alignment because of its expected high cost, weaker business case and impacts on the Peak District National Park.

Why a New Line?

The existing Hope Valley route faces several constraints:

- numerous sharp curves
- Victorian tunnels
- mixed passenger and freight operation
- environmental constraints within the Peak District
- limited opportunities for sustained 125 mph running

Government evidence notes that even with ongoing upgrades, the emphasis is on **capacity improvements** and selective speed enhancements rather than creating a new high-speed railway.

Strategic Vision

Create:

Northern Alpine Railway

A purpose-built railway linking

- Sheffield Victoria
- Deepcar
- Penistone (optional parkway)
- Glossop/Woodhead (new tunnel)
- Manchester Piccadilly

Core Specifications

Item	Specification
Maximum speed	125 mph (200 km/h)
Electrification	25kV AC
Signalling	ETCS Level 2/3
Double track	Entire route
Design life	120 years

Route Length

Approximately

65 km

New build:

45 km

Existing upgraded corridor:

20 km

Sheffield Victoria

The proposal includes constructing a new Sheffield Victoria station.

This is not without precedent.

During HS2 planning, Sheffield Victoria was extensively assessed as a possible city-centre high-speed station. HS2 engineers concluded that a through alignment at **200 km/h** was technically achievable, although it would require substantial reconstruction and widening of the corridor.

Capital Cost Estimate

Component	Cost
New railway	£3.2bn
Woodhead base tunnel	£2.8bn
Sheffield Victoria station	£600m
Manchester approaches	£900m
Electrification	£500m
Digital signalling	£300m
Rolling stock	£700m
Land, planning & contingency	£2.0bn

Total Programme

£11.0 billion

This is an indicative strategic estimate. Actual costs would depend on route selection, tunnelling extent, environmental mitigation and procurement.

Journey Times

Route	Current	Proposed
Sheffield–Manchester	50–54 min	24–28 min
Sheffield–Manchester Airport	~75 min	~40 min
Sheffield–Liverpool	~95 min	~60 min

Passenger Forecast

Year	Annual Journeys
Opening	12 million
Year 10	18 million

Year	Annual Journeys
Year 20	23 million

Fare Revenue

Average fare

£12

Passengers

18 million

Annual revenue

£216 million

Property Development

Major regeneration hubs:

Sheffield Victoria

Potential:

- 5,000 apartments
- offices
- hotels
- university expansion
- retail

Development value:

Approximately

£2.5 billion

Deepcar

Transit-oriented town expansion

2,000 homes

Woodhead Gateway

Tourism

Hotel

Conference centre

Outdoor recreation

Manchester East

Commercial redevelopment

Economic Impact

Improved connectivity between Sheffield and Manchester could:

- enlarge effective labour markets,
- reduce business travel times,
- attract investment,
- support advanced manufacturing,
- improve university collaboration.

The Government has consistently argued that better east–west connectivity is central to raising productivity across Northern England.

Illustrative Wider Economic Benefits

Source	Annual Impact
Labour productivity	£900m
Business investment	£700m
Tourism	£300m
Property uplift	£450m
Manufacturing	£600m
Innovation	£350m

Total

Approximately £3.3 billion per year

These are scenario-based estimates of wider economic activity rather than official forecasts. A formal Green Book and Department for Transport appraisal would be required.

Construction Employment

Peak construction workforce

15,000

Supply chain jobs

20,000

Permanent jobs

5,000+

Environmental Benefits

Potential benefits include:

- reduced reliance on car travel across the Pennines,
- greater rail mode share,
- relief for the existing Hope Valley line, allowing more freight and stopping services.

However, a new line through or beneath the Peak District would also present significant environmental challenges, including impacts on protected landscapes, habitats and planning constraints.

Funding Structure

Source	Investment
UK Government	£6.5bn
National Infrastructure Finance	£2.0bn
Bank of Leeds	£1.0bn
Pension Funds	£1.5bn

Financial Returns

Metric	Value
Capital Cost	£11bn
Annual Fare Revenue	£216m
Estimated Wider Economic Impact	£3.3bn/year
New Homes Enabled	7,000+
Construction Jobs	35,000 job-years

Strategic Assessment

Strengths

- Creates a true high-speed Trans-Pennine corridor.
- Re-establishes Sheffield Victoria as a regional rail gateway.
- Frees capacity on the Hope Valley line for local passenger services and freight.
- Supports long-term regeneration around Sheffield Victoria.

Risks

- Very high capital cost.
- Complex engineering, particularly if extensive tunnelling is required.
- Significant environmental and planning challenges within the Peak District National Park.
- The Government has previously concluded that a comparable new alignment did not offer sufficient value for money relative to upgraded existing routes.

Recommendation

From a purely strategic perspective, a **new Sheffield Victoria–Manchester railway** would provide faster journeys and greater long-term network capacity than incremental upgrades to the Hope Valley Line.

However, based on current public evidence, the strongest business case is likely to be a **phased approach**:

1. Reopen Sheffield Victoria as a regional station.
2. Complete Hope Valley electrification and capacity upgrades.
3. Protect the former Woodhead corridor where feasible.
4. Undertake a fresh strategic study into a future high-speed Trans-Pennine route if demand, population growth and economic conditions justify it.

This approach aligns more closely with existing transport policy while preserving the option of a transformational new corridor in the future.